ANLP | Wordsmiths

DiffusEdit

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Abstract

Reference articles go out of date as events develop, and updating them by hand is slow. Regenerating the article with a fine-tuned model trains every parameter and leaves no record of which tokens changed. DiffusEdit instead edits the article in place. The frozen diffusion model rewrites only the stale positions, and the system repeats the process on sentences that the rewrites themselves made wrong.

Literature

- **FRUIT (Logan IV et al., 2022)**
  - Defines our task: update a whole article from new evidence.
  - Its EdiT5 model rewrites the article left to right, copying unchanged sentences through special tokens.
- **Detect, Remask, Repair (DRR) (Zou et al., 2026)**
  - Applies a diffusion model to existing text plus external evidence.
  - Detect: scores each token of a stale summary with [MASK]-DISC, a linear head over the diffusion model's hidden states.
  - Remask and repair: masks the highest-scoring tokens and infills each one with exactly one new token, then repeats, re-checking against the same evidence each round.
  - Gap: fixed-length edits, no tracking of cascading edits, and focuses only on small summaries.

Problem

- **Reference text ages rapidly:** Real-world events (news, Wikipedia) constantly makes existing facts out-dated.
- **Current LLM solutions are brute-force:** Trains every parameter of the model and keeps no record of which tokens changed.
- **The "Cascading Edit" Dilemma:** Facts are logically connected. If you change a fact in paragraph one, sentences in paragraph three might suddenly become completely unsupported.

Example:

Sentence 1: “Three people died”

Sentence 2: “The small death toll limited the political fallout“

A week later, let’s say the death toll increases to 42.

S1: wrong
S2 : wrong by dependence

Contributions

- We introduce and release **Wiki Wrong**, **15,346** Wikipedia leads with sentence labels, stale spans, gold rewrites and cascade links, with every edit reversed.
- A linear **sentence classifier** and a learned attention based **token staleness classifier** over a frozen **LLaDA** model**.**
- The **infilling rules** (variable length, evidence copies, edit penalty, rejection, evidence veto, span cap) and a learned **ranker**.
- A loop of up to **three** rounds with the consequence classifier, compared against staleness delta and edited as evidence.

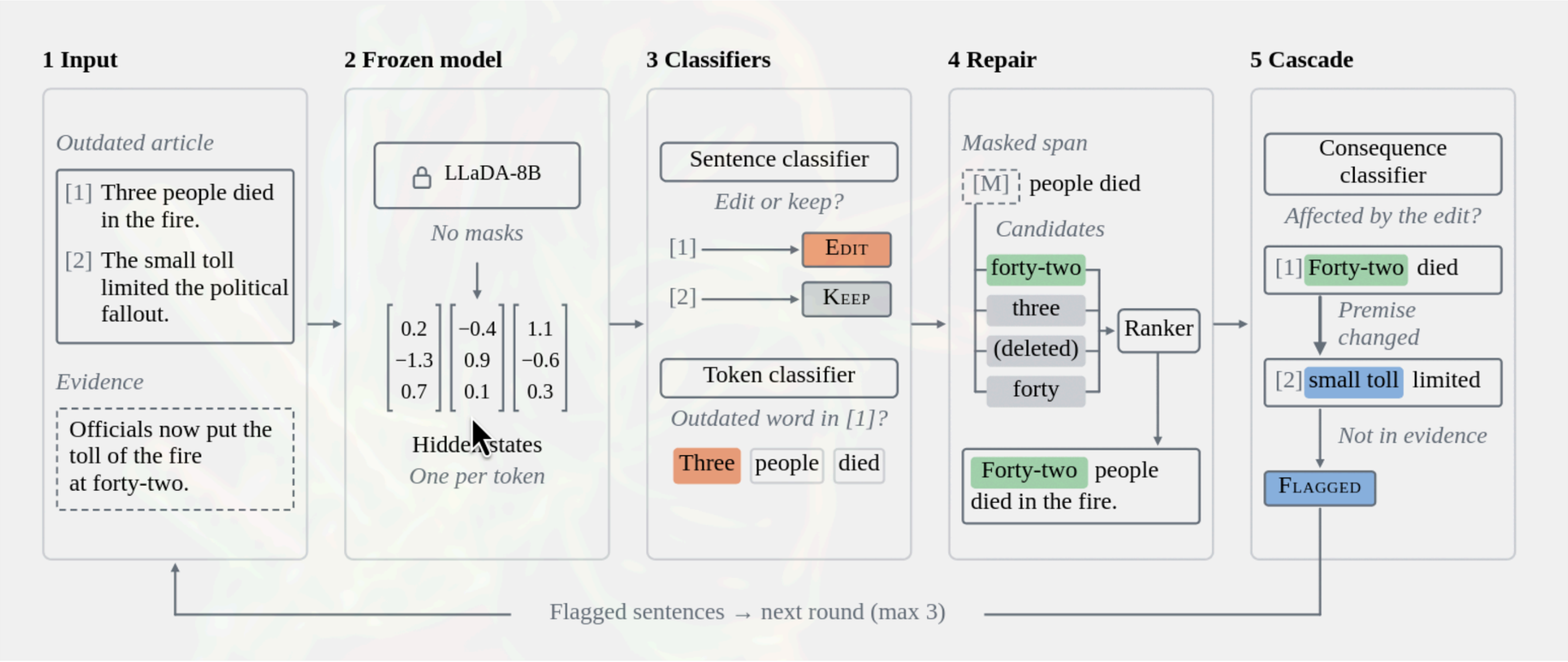
Limitations of FRUIT & Label Noise

- **The Noise Problem:**
  - Real Wikipedia edits mix factual updates with surface rewrites (active ↔ passive, word substitutions).
  - Text-alignment heuristics misclassify stylistic changes as EDIT or DROP.
- **Human Audit Verification:**
  - Audited 100 sentences across 23 training articles (inter-annotator $\kappa = 0.60$--0.86).
  - 13.1% label error rate (Wilson 95% CI: 7.8%–21.2%).
  - 100% of errors were false EDIT labels on unchanged facts (8.1% purely stylistic).
  - 22.6% of all training EDIT sentences change neither an entity nor a number.
- **Further problems:**
  - Editors also move sentences out of the lead, with no evidence behind the change.
  - Only 57 of 338 FRUIT rewrites that add words have those words in the source or the evidence.

Wiki Wrong

- **Deterministic Supervision:**
  - Starts from clean Wikipedia introductions; generates exact character-level labels for KEEP, EDIT, and DROP (retraction notices).
  - Injects cascading dependencies ($s_j \rightsquigarrow s_k$): updates where s_k is contradicted without direct evidence.
- **Reversed Formulation (Anti-Memorization):**
  - Original Wikipedia text = Outdated Source.
  - Synthetic counter-factual = Target Update + Evidence.
  - A model that reproduces a memorised fact is wrong, so the gold update can be reached only through the evidence.
- **Dataset Scale:**
  - 12,180 Train / 1,564 Val / 1,602 Test articles.
  - Evaluated on 901 supported articles (1,670 edit sentences) and 220 linked articles (cascade test set).

Pipeline


1 Input
Outdated article
[1] Three people died in the fire.
[2] The small toll limited the political fallout.
Evidence
Officials now put the toll of the fire at forty-two.
2 Frozen model
LLaDA-8B
No masks
0.2 −1.3 0.7
−0.4 0.9 0.1
1.1 −0.6 0.3
Hidden states
One per token
3 Classifiers
Sentence classifier
Edit or keep?
[1] EDIT
[2] KEEP
Token classifier
Outdated word in [1]?
Three people died
4 Repair
Masked span
[M] people died
Candidates
forty-two
three
(deleted)
forty
Ranker
Forty-two people died in the fire.
5 Cascade
Consequence classifier
Affected by the edit?
[1] Forty-two died
Premise changed
[2] small toll limited
Not in evidence
FLAGGED
Flagged sentences → next round (max 3)

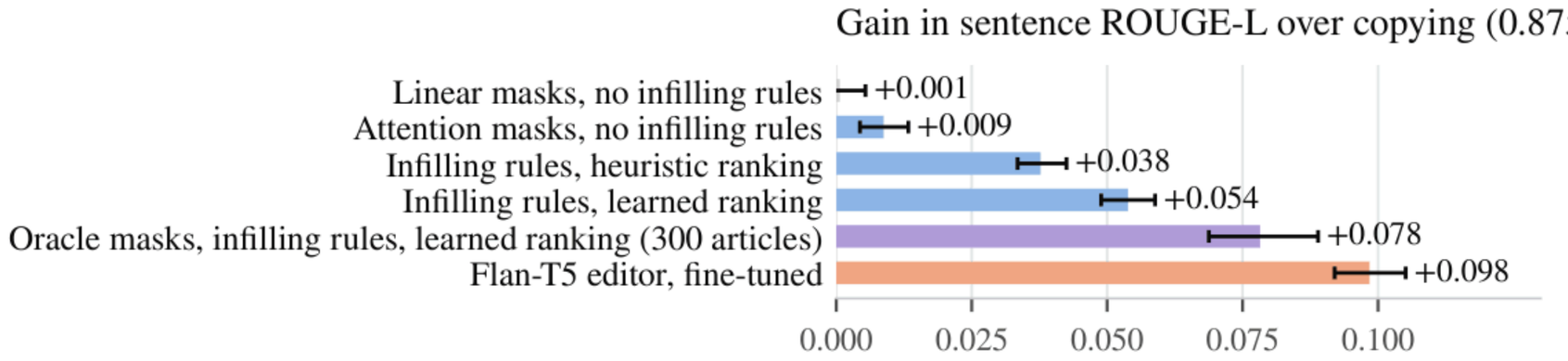
Evaluation

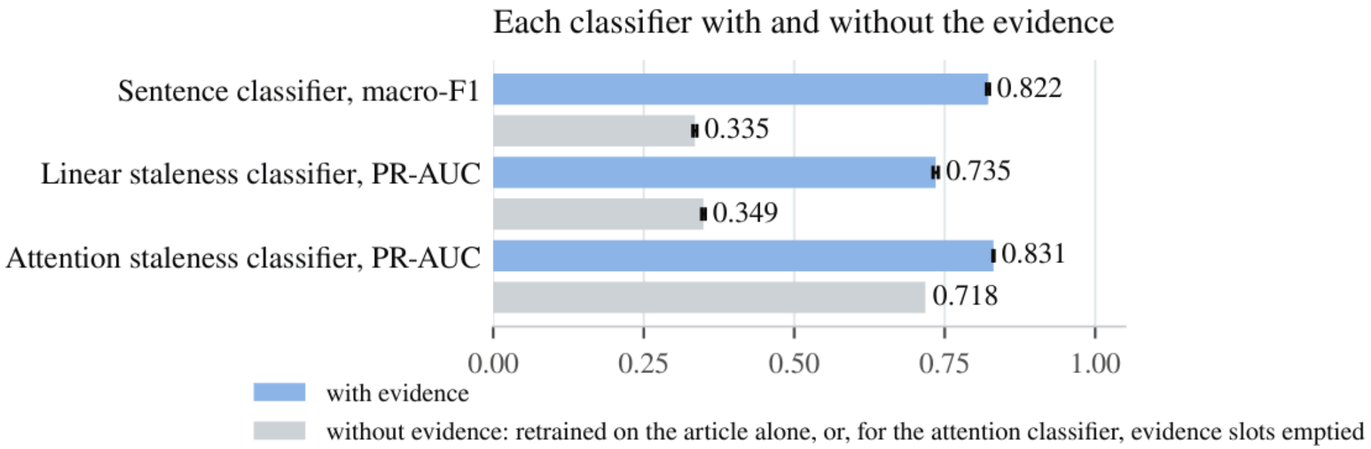
- **Document & Multi-Round Cascade Tracking:**
  - **Cascade Recall:** Percentage of secondary, evidence-absent errors successfully flagged for repair.
  - **Cascade ROUGE-L:** Quality of repairs measured specifically on secondary cascade sentences.
  - **UpdateROUGE-L:** Document-level ROUGE-L restricted strictly to modified sentences (avoids score inflation from untouched text).
  - **KEEP Verbatim:** Percentage of clean, factual sentences preserved completely untouched.
- **Classifier scores:**
  - **Sentence Classifier:** Macro-F1 across KEEP, EDIT, and DROP (3-class).
  - **Staleness Classifier:** PR-AUC over tokens in EDIT sentences (handles high class imbalance: only 11.3% of tokens are stale).
  - **Consequence Classifier:** AUROC on dependency sentence pairs (s_j, s_k).

Evaluation

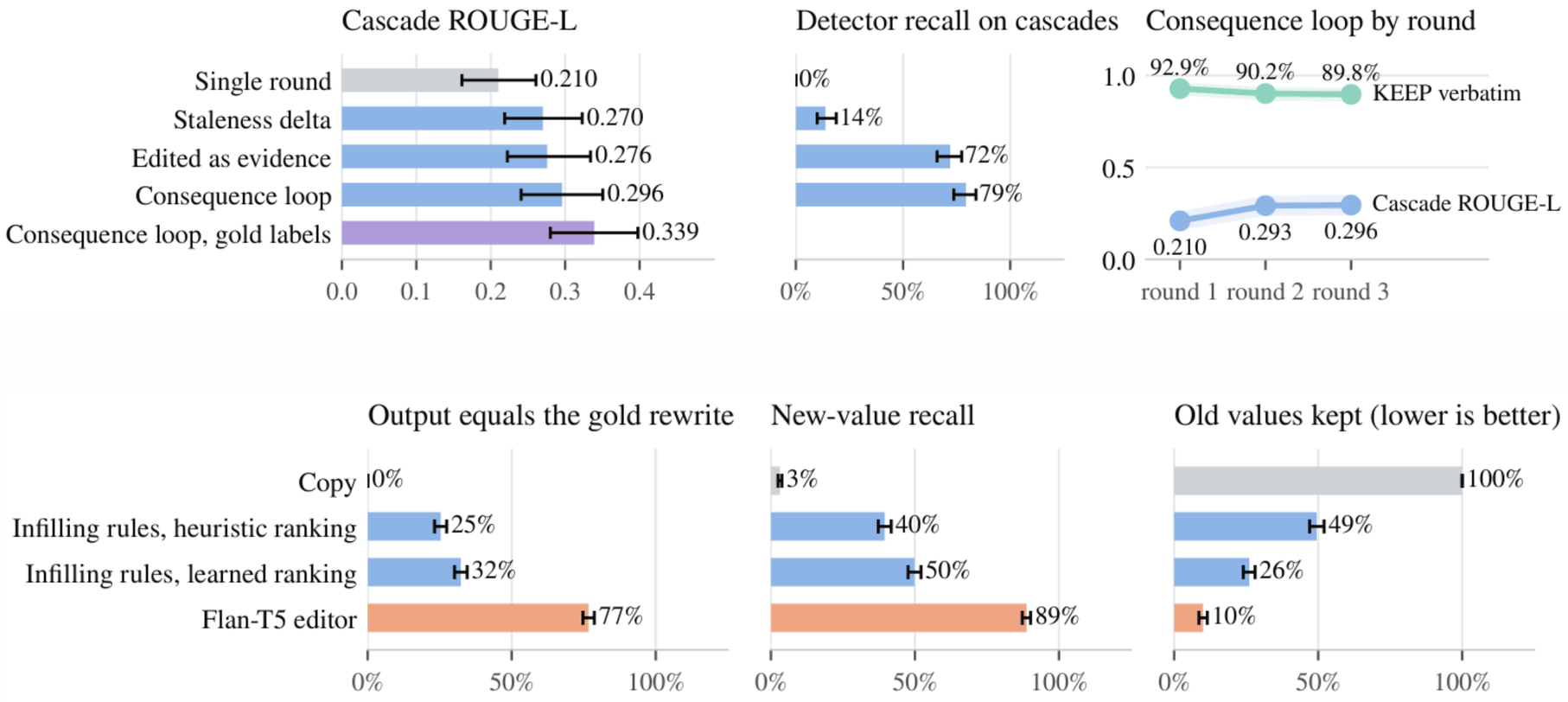
- **Sentence-Level Infilling Quality:**
  - **ROUGE-L (F1):** Overlap between the repaired sentence and the gold update (Δ over Copy baseline of 0.875).
  - **Exact Match (EM):** Percentage of sentences matching the gold target update 100% verbatim.
  - **New-Value Recall:** Share of new values that occur in the output as whole words.
  - **Old-Value Kept (↓):** Share of old values that still occur in the output.

Results


Gain in sentence ROUGE-L over copying (0.875)
Linear masks, no infilling rules
Attention masks, no infilling rules
Infilling rules, heuristic ranking
Infilling rules, learned ranking
Oracle masks, infilling rules, learned ranking (300 articles)
Flan-T5 editor, fine-tuned
+0.001
+0.009
+0.038
+0.054
+0.078
+0.098
0.000
0.025
0.050
0.075
0.100


Each classifier with and without the evidence
Sentence classifier, macro-F1
0.822
0.335
Linear staleness classifier, PR-AUC
0.735
0.349
Attention staleness classifier, PR-AUC
0.831
0.718
0.00
0.25
0.50
0.75
1.00
with evidence
without evidence: retrained on the article alone, or, for the attention classifier, evidence slots emptied

Results


Cascade ROUGE-L
Single round
Staleness delta
Edited as evidence
Consequence loop
Consequence loop, gold labels
0.210
0.270
0.276
0.296
0.339
0.0
0.1
0.2
0.3
0.4
Detector recall on cascades
0%
14%
72%
79%
0%
50%
100%
Consequence loop by round
1.0
0.5
0.0
92.9%
90.2%
89.8%
KEEP verbatim
Cascade ROUGE-L
0.210
0.293
0.296
round 1
round 2
round 3
Output equals the gold rewrite
Copy
Infilling rules, heuristic ranking
Infilling rules, learned ranking
Flan-T5 editor
0%
25%
32%
77%
0%
50%
100%
New-value recall
3%
40%
50%
89%
0%
50%
100%
Old values kept (lower is better)
100%
49%
26%
10%
0%
50%
100%

Results

Articles	System	Upd. R-L	Ent. P	Ent. R	KEEP verb.	Unsupp. ↓	Casc. R-L
300 test	Single round	0.476	0.675	0.382	92.7%	**0.19**	
	Consequence loop	0.490	0.665	0.399	90.7%	0.20	
	Flan-T5 editor	**0.911**	**0.915**	**0.885**	**97.9%**	0.25	
	Loop, no evidence	0.062	0.116	0.015	93.7%	0.09	
220 linked	Single round	0.436	0.866	0.330	92.9%	**0.30**	0.210
	Consequence loop	0.470	0.847	0.360	89.8%	0.35	0.296
	Flan-T5 editor	**0.926**	**0.949**	**0.890**	**95.6%**	0.98	**0.815**

Table 1: Article scores on a seeded sample of 300 test articles and on the 220 test articles with a cascade link. Ent. P and Ent. R are the precision and recall of the mentions of the changed sentences against the gold update, with precision averaged over the articles in which a system changed a sentence. Copy scores 0 UpdateROUGE-L, 100% KEEP verbatim and 0 unsupported mentions. Bold marks the best system in each group, leaving out the no-evidence control, which changes almost nothing.

Setting	ROUGE-L	EM	Changed
Fit on 500 training articles, 300 ablation articles			
Copy	0.876	0%	0%
Classifiers, heuristic ranking	0.908	24%	58%
Classifiers, refit learned ranking	**0.917**	**29%**	72%
Flan-T5 editor	0.875	2%	4%
Fit on all training articles, same 300 articles			
Classifiers, learned ranking	0.930	33%	83%
Flan-T5 editor	**0.979**	**76%**	93%
Missing-value stress set, 200 validation articles			
Copy	0.882	0%	0%
Without infilling rules	0.827	5%	81%
With infilling rules	0.883	9%	41%
Flan-T5 editor	**0.926**	**38%**	47%
Old-value stress set, 200 validation articles			
Copy	0.882	0%	0%
Without infilling rules	0.878	13%	80%
With infilling rules	0.878	0%	9%
Flan-T5 editor	**0.951**	**61%**	86%

Table 5: Sentence scores of the matched-budget comparison and of the stress sets. In the matched comparison the editor uses the validation split for early stopping and the classifiers a reranker refit on validation pools of the 500-article staleness classifier. Changed is the share of EDIT sentences whose output differs from the source. Bold marks the best ROUGE-L and EM in each group.

Limitations

- **Inference Bottleneck:**
  - DiffusEdit requires 9.1 s per article (due to iterative masking and multi-candidate PLL scoring) compared to 0.35 s per article for the fine-tuned T5 editor
- **Synthetic data only:**
  - Every edit replaces one value by another of the same type. Rewordings, edits that change meaning without changing a named value, and the noise of human revisions are absent, so a result on the benchmark does not transfer to real revision data until it is checked there.
- **Training Seeds:**
  - The repair and loop results come from one attention staleness classifier and one consequence classifier each, and the reported intervals reflect the sampling of test articles and not the training seed.
- **Evidence Dependency:**
  - The rules assume the evidence states every new value and no old value. With the new values removed for half the edits the gain over copy falls to zero, and with an added passage holding the old value the rules score below copy.

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Questions?

Your doubt is [MASK] - we'll infill it

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THANK YOU

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